How Hackable Is Your Speech Quality Metric? A Corrected Protocol, a Benchmark, and What Patching Buys

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***Abstract*—Speech quality predictors are increasingly used as rewards, yet no agreed measure of their hackability exists. The usual measurement has two flaws. First, the perturbation reaches the predictor through a processing chain — here a neural codec — that shifts the score on its own, which scoring against the raw input charges to the attack. Referencing the unperturbed round trip instead changes measured hackability by up to a factor of four (0.31 to 0.08 for one defence). Second, one trained attacker is a sample, not a measurement: five attackers differing only in random seed reach success rates from 0.00 to 0.38 against one fixed predictor, so a defence claim needs the worst case over several. Under this protocol, four published predictors differ widely: NISQA is hacked on 90% of utterances, SSL-MOS on 21%, DNSMOS on 14% and UTMOS on 6%. We then audit a closed attack–detect–patch loop. It hardens the predictor only in its own attack space, by less than the spread between attackers; a random-perturbation baseline matches it; and it costs up to 0.30 system SRCC out of domain. Enhancers post-trained against patched predictors hack them far less (PESQ -0.03 versus -0.23). Code, preregistration and run outputs are released.**

***Index Terms*—speech quality assessment, reward hacking, reinforcement learning, adversarial robustness, MOS prediction**

I. Introduction

Automatic mean-opinion-score (MOS) predictors are both the standard evaluation instrument for speech generation [1], [2] and, increasingly, the *reward* that trains it [3]–[5]. Optimising such a metric degrades the perceptual quality it was built to measure: an enhancer tuned for PESQ reaches a state-of-the-art 3.82 PESQ yet scores very poorly in a listening test [6]. The predictors can also be attacked directly. An attack adds a small perturbation — to the waveform, or to a neural codec's latent representation of it — and asks how far the predicted score moves; a white-box attack shapes it with the predictor's gradients, a black-box attack sees only the returned score. White-box attacks on UTMOS start from clean speech, degrade it severely, and still hold its predicted MOS at the clean value [2]: a high predicted score is no evidence of high quality. What such an attack does not do is push the score *above* the score of the speech it started from — and that is exactly what a generator trained against the predictor is rewarded for, since every point it adds is reward it collects. A practitioner choosing which predictor to reward with therefore has no evidence to go on: predictors are picked for their published correlation with human ratings, for precedent, or for what a toolkit ships — none of which says how much score an optimiser can extract without improving the audio.

There is no protocol for measuring how hackable a predictor is, and no study that puts the predictors people actually use in front of the *same* attacker, at the *same* perturbation strength, under the *same* rule for what counts as a hack. Published attacks differ in all three. Lin et al. [7] drive DNSMOS to a fixed target score under a $-30\,\mathrm{dB}$ peak bound and treat any change in the prediction as an attack; EventTrojan [8] poisons 3% of a predictor's training data with an inserted 0.2–$0.5\,\mathrm{s}$ dropout and counts a score landing within 0.5 MOS of the attacker's target; the UTMOS attacks [2] impose no norm bound at all and report a PESQ–UTMOS trade-off over 30 utterances. Their numbers therefore do not share an axis: a predictor can look robust because it is, or merely because it was attacked more gently.

The same failure sits in every pipeline that puts a learned quality model inside an optimisation loop. Speech enhancement is now routinely post-trained against DNSMOS or NISQA rewards [4], [9]: any perturbation that raises the reward without improving the speech is a shortcut, and reinforcement learning (RL) is a search that finds shortcuts, so a deployed system can degrade on the very axis it reports improving. Text-to-speech and voice conversion use the same predictors to filter training corpora [10], [11], pick checkpoints and hyper-parameters [12], [13] and rank challenge submissions [14], so a hackable predictor quietly promotes the wrong system. In hearing aids these predictors stand in for listeners who cannot easily A/B test their own device, so a metric an inaudible perturbation lifts by a full MOS point gives no assurance that a fitting which scores better sounds better. In each case the practitioner's only lever is which predictor to trust, and that choice is currently made blind.

This paper supplies the missing protocol and the missing comparison. A hack is a score that rises *because of the perturbation*, so the rule's reference is the processing chain's own output, not the raw input, which folds in how the predictor scores the codec itself (-0.10 to $+0.14$ MOS across predictors, up to $+0.95$ across defences). And because an attacker is itself trained, one attacker is one sample: five attackers differing only in random seed span success rates from 0.00 to 0.38 against one fixed predictor. Fixing both, we benchmark four

widely used predictors at one perturbation size and find a spread of more than an order of magnitude, the most hackable — NISQA, commonly used as an RL reward — hacked on 90% of utterances.

We then apply the protocol to the defence the diagnosis suggests: a closed attack–detect–patch loop, the speech counterpart of Adv-RM [15], whose modality gap we cannot cross to run it as a baseline. Correctly measured, it hardens only the space it trains on, by less than the spread between attackers, and a baseline with no RL and no detector matches it.

Contributions. (i) A measurement protocol — reference the hack rule to the processing chain's output, report the worst case over k trained attackers — with evidence that both corrections change conclusions (Sec. V). (ii) The first benchmark of common predictors at one perturbation size and one detector (Table I, Fig. 2a). (iii) An audit of closed-loop patching: narrow hardening, negative transfer, a trivial baseline that matches it, and the out-of-domain cost (Table II). (iv) Downstream evidence on reward hacking of a small enhancement model (Table III). All data, models and analysis scripts are released.

II. Related work

Quality metrics can be gamed. Beyond the PESQetarian [6], FlowSE-GRPO observes implausible DNSMOS gains under single-metric optimisation [4], and composite rewards are the usual remedy [5], [9]. Asonitis et al. [1] diagnose hacking — the optimised score rises as an independent panel falls — but leave the predictor untouched.

Attacking quality predictors. None of the attacks of Sec. I proposes a defence. Lin et al. [7] adversarially train DNSMOS once against waveform perturbations, without a hack detector or a downstream generator; RRPO [16] hardens a TTS reward model in one pass, as does gradient-norm regularisation for image quality models [17].

Closing the loop. Adv-RM [15] applies the same recipe to LLM reward models. Related ideas alternate policy and reward-model updates [18]–[20], ensemble reward models [21], [22], harden agent benchmarks [23] or post-train adversarially for music [24].

III. Method

Let R be a mean-opinion-score (MOS) predictor mapping a waveform to a scalar. The loop (Fig. 1) has three parts.

Attacker. Speech x is encoded to a continuous latent $z = \mathrm{Enc}(x)$ by a frozen neural codec [25]. A policy π_θ proposes an action $a \sim \mathcal{N}(\mu_\theta(z), \sigma^2)$ in per-channel standardised latent units; the perturbation $\delta = \mathrm{expand}(a)$ is rescaled to a fixed root-mean-square size ϵ, and the attacked signal is $y = \mathrm{Dec}(z + \delta \odot s_z)$ with s_z the per-channel latent standard deviation. The policy is trained with group relative policy optimisation (GRPO) [26]: for each utterance we draw G actions, score them with the frozen target predictor, and use group-normalised advantages $A_{ij} = (r_{ij} - \bar{r}_i)/(\mathrm{std}_j r_i + \eta)$ with $r = R(y)$, optimised with a clipped surrogate. The action is time-invariant (one C-dimensional offset per utterance), which keeps the policy-gradient variance tractable: a frame-rate action space did not learn at any size we tried. The attacker needs only forward scores, so any predictor can be a target.

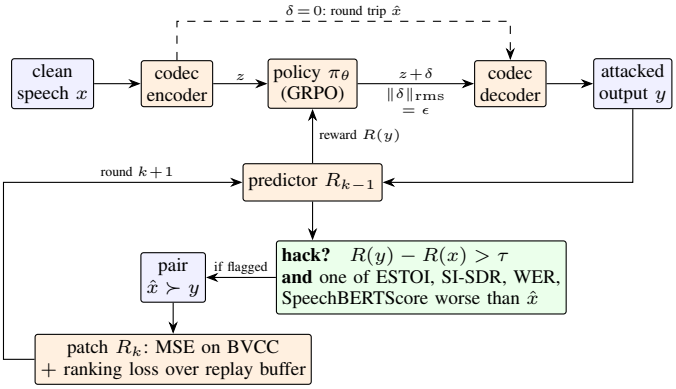


Fig. 1. The attack–detect–patch loop. A GRPO policy perturbs codec latents at a fixed size; an output is flagged only if its score rises above that of the zero-perturbation round trip $\hat{x}$ (dashed) *and* an independent panel degrades relative to $\hat{x}$. Flagged pairs patch the predictor; the next round retrains the attacker.

Label-free hack detection. An attacked output y is flagged as a *hack* when both rules hold: (i) *ceiling rule* — the predictor scores the attacked signal more than τ MOS above the codec round trip $\hat{x} = \mathrm{Dec}(z)$, i.e. $R(y) - R(\hat{x}) > \tau$; and (ii) *panel rule* — at least one independent measure is worse for y than for $\hat{x}$ by more than a per-measure margin: ESTOI, SI-SDR, SpeechBERTScore [27] (all against x), or the word error rate of Whisper transcripts of y against those of x. Referencing both rules to $\hat{x}$ rather than to the raw input x is what isolates the effect of the perturbation from the codec's own round-trip loss; the loop as run used x in the ceiling rule, and every number we report is recomputed against $\hat{x}$ (Sec. V). No human labels are used. The fraction of utterances an attacker gets flagged this way is its *attack success rate* (ASR), our measure of hackability.

Patch. Every flagged hack contributes a preference pair $\hat{x} \succ y$, and the predictor is fine-tuned on its original regression loss plus a margin ranking loss $\max(0, m - (R(\hat{x}) - R(y)))$ over a replay buffer holding every previous round's hacks. Using $\hat{x}$ rather than x as the preferred side prevents a "codec artefact" shortcut; an exploratory ablation ($n=1$) confirms it (in-loop ASR 0.010 by round 3 against 0.172 with x). The three steps repeat for K rounds, giving $R_1, \ldots, R_K$.

Multi-space variant. The same loop can draw its pairs from several attack spaces at once. In addition to the codec latent space above we use (i) a second codec (DAC [28], quantised latent) and (ii) white-box waveform PGD that maximises R under $\|\delta\|_\infty \le 0.002$. All flagged hacks share one buffer.

IV. Experimental setup

Predictor and data. R_0 is SSL-MOS [29]: a wav2vec 2.0 base encoder with mean pooling and a linear head, trained with MSE on the BVCC [30] training split (4 974 utterances) for 8 000 steps (AdamW, $2{\times}10^{-5}$ encoder / 10^{-3} head, batch 8, 6 s crops), selecting the best validation-MSE checkpoint. Human agreement is evaluated on 13 test sets: BVCC test, the Voice-MOS'22 out-of-domain set [31], [32], three NISQA sets [33],

TABLE I
HACKABILITY OF COMMON PREDICTORS AT ONE SIZE ($\epsilon = 0.5$, ENCODEC LATENTS), FIVE INDEPENDENTLY TRAINED ATTACKERS EACH, 500 BVCC-TEST UTTERANCES; "WORST" IS THE LARGEST OF THE FIVE.

Predictor	offset	mean ASR	worst ASR	mean gain
SSL-MOS (BVCC)	-0.10	0.205	0.252	+0.191
UTMOS22	-0.07	0.057	0.076	+0.144
NISQA	+0.14	0.896	0.916	+0.954
DNSMOS OVRL	+0.02	0.136	0.200	+0.124

SOMOS [34], TMHINT-QI [35], four VoiceMOS'23 tracks [36] and the URGENT'24/'25 quality-assessment sets [37], [38].

Attacker and detector. The attack space is the continuous EnCodec-24 kHz latent (128 channels at 75 Hz); the policy is a four-layer 1-D convolutional network (256 hidden units) with a per-channel learned σ, trained for 2 000 GRPO steps ($G = 8$, batch 8, 3 s crops, lr 10^{-3}, clip 0.2, two inner epochs). $\epsilon = 0.5$ was selected on validation alone (Sec. V). The detector uses $\tau = 0.3$ MOS with margins of 0.02 ESTOI, 1 dB SI-SDR, 0.02 SpeechBERTScore and any WER increase (Whisper-small; HuBERT-base layer 8).

Patching and baselines. Each round fine-tunes for 1 500 steps with equal MSE and ranking batches (margin $m = 0.5$, weight 1) and $K = 3$ rounds. Three defences share the 4 500-step budget and loss, differing in where the rejected side comes from: Gaussian noise pairs (SNR 5–30 dB), waveform-PGD pairs, and random latent perturbations at the same ϵ (no RL, no detector). PGD uses $\|\delta\|_\infty \leq 0.002$ with 50 steps at evaluation but 10 steps inside the multi-space loop and 5 in the PGD-pair baseline, so those defences meet a weaker attacker than they are tested against; training is step-matched but not pair-matched (replay buffer 5.8k to 8.9k pairs against 19.9k random pairs, favouring the baseline).

Downstream (E3). A 2.6 M-parameter STFT mask enhancer is pre-trained on VoiceBank-DEMAND and post-trained with GRPO for 2 500 steps, the policy adding per-frame offsets on 32 ERB bands with no KL penalty, so only the reward model limits hacking. Evaluation uses the 824-utterance test set.

Protocol. Seeds, primary metrics and decision rules for E1–E3 were preregistered before any test-label contact, with one amendment (attack action space and the ϵ rule) made on validation evidence before the main runs; the multi-space variant and the ablations are exploratory, and the corrected ceiling reference is a post-hoc re-analysis of stored outputs. Results are mean $\pm$ sd over n seeds.

V. RESULTS

The reference of the hack rule changes the answer. A predictor that scores codec-processed audio differently from raw audio shifts every raw-referenced measurement with it, and that offset is neither negligible nor constant: -0.10 (SSL-MOS), -0.07 (UTMOS), $+0.02$ (DNSMOS) and $+0.14$ (NISQA) across predictors (Table I), and up to $+0.95$ across the defences of Table II. The consequences are concrete: under the naive reference DNSMOS looks *more* hackable than it is (0.183 vs 0.136), NISQA *less* (0.839 vs 0.896), and the multi-space defence below flips from apparently catastrophic (0.306) to among the most robust (0.082).

One attacker is not a measurement. Retraining the attacker with different seeds against one fixed predictor gives ASR from 0.000 to 0.261 on R_0 and 0.000 to 0.382 on R_3 (sd 0.130 and 0.180) — as large as every defence effect we measure. Between *predictors*, by contrast, the spread is small relative to the differences (NISQA 0.876–0.916 against UTMOS 0.042–0.076), so the benchmark ranking is solid where defence comparisons are not. We therefore report the worst of five attackers.

Common predictors differ by more than an order of magnitude (Table I). At one perturbation size, NISQA is hacked on 89.6% of utterances for a mean gain of $+0.954$ MOS, SSL-MOS on 20.5%, DNSMOS OVRL on 13.6% and UTMOS on 5.7%; worst-case rates are 0.916, 0.252, 0.200 and 0.076. As NISQA and DNSMOS are routinely used as RL rewards, the choice of reward is a first-order robustness decision, and the ranking is not predictable from in-domain accuracy.

Patching hardens narrowly, and a trivial baseline matches it (Table II). Three rounds against the EnCodec attacker cut ASR from 0.226 ± 0.205 (R_0) to 0.050 (R_1) and 0.017 (R_2), and the replayed round-1 attacker never succeeds. The effect is not monotone — $K = 3$ was preregistered rather than selected, and R_2 dominates R_3 on both axes — and the $R_0 \to R_3$ difference does not hold its sign across seeds (-0.073, -0.039, $+0.379$), so our preregistered H1, which passes under the naive reference, fails under the corrected one. Pairs from *random* latent perturbations at the same size, with no RL and no detector, give the lowest in-space ASR of any defence (0.019 ± 0.016). Since the patch margin ($m = 0.5$) exceeds the detection threshold ($\tau = 0.3$) we swept $\tau \in [0.1, 1.0]$: the ordering is unchanged, though R_3 alone keeps a tail of large hacks (ASR 0.045 at $\tau = 1.0$; 90th-percentile score gap 0.651 vs 0.359 for R_0).

No transfer, and one real trade. Against an unseen DAC attacker the EnCodec-patched predictors are *worse* than undefended (0.065 to between 0.13 and 0.16), and white-box PGD succeeds on nearly every utterance for every EnCodec-trained defence. Only multi-space training changes this: it resists DAC (0.016) and is the one predictor PGD does not fully break (0.841 ± 0.089), paying with the $+0.95$ MOS reconstruction offset and the largest loss of human agreement. In-loop, DAC and PGD fall round by round while the EnCodec attacker stays flat (Fig. 2b).

Fidelity: ranking holds in domain, calibration and out-of-domain agreement do not. The patched predictors keep their BVCC system ranking (R_3 0.920 vs R_0 0.922, meeting the preregistered -0.02 margin) but utterance MSE rises $0.234 \pm 0.011 \to 0.272 \pm 0.022$, a calibration loss that matters for a reward model. Out of domain the mean system SRCC falls from 0.731 to 0.690, 0.682 and 0.637 over rounds one to three and to 0.552 for R_3^{multi}, worst on noisy speech (TMHINT-QI, where the R_0 and R_3 bootstrap intervals do not overlap; TMHINT-QI and VoiceMOS'23 track 3 share a condition grid

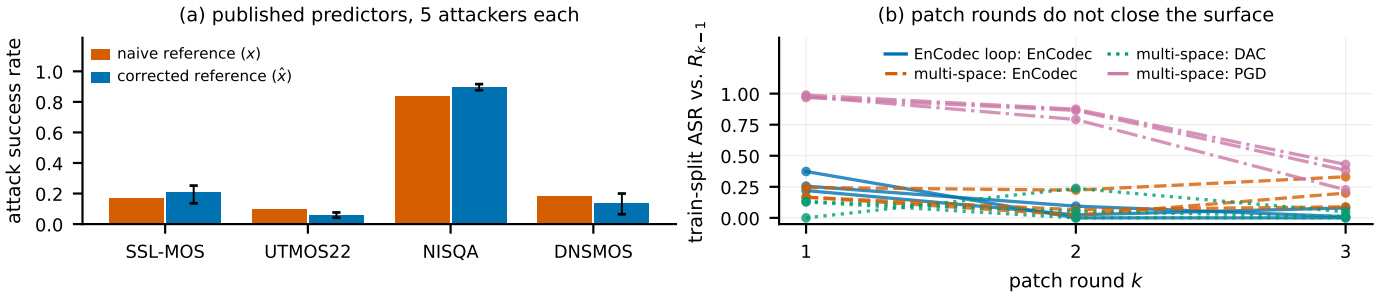


Fig. 2. (a) Benchmark of four common predictors at one perturbation size under the naive (x) and corrected ($\hat{x}$) ceiling references; error bars span the five attackers, and the reference moves DNSMOS and NISQA in *opposite* directions. (b) In-loop attack success rate (BVCC train, corrected reference) of the round-k attacker against the predictor patched in round $k-1$, one line per seed: the single-space loop converges in its own space; under multi-space training DAC and PGD fall while the EnCodec attacker stays flat ($+0.01$).

TABLE II

AUDITING THE LOOP. CORRECTED-REFERENCE ASR PER ATTACK SPACE (FRESH ATTACKER, BVCC TEST) AND AGREEMENT WITH HUMAN MOS (BVCC SYSTEM SRCC; OUT-OF-DOMAIN MEAN OVER 12 SETS), MEAN±SD OVER 3 SEEDS.

	offset	ASR by attack space			human agreement	
Predictor	$R(\hat{x})-R(x)$	EnCodec	DAC	PGD	BVCC	OOD
R_0 (no defense)	-0.09±0.02	0.226±0.205	0.065±0.113	0.999±0.002	0.922±0.004	0.731±0.006
R_1	-0.06±0.03	0.050±0.048	0.161±0.033	1.000±0.001	0.911±0.009	0.690±0.014
R_2	0.00±0.07	0.017±0.006	0.141±0.025	1.000±0.000	0.918±0.003	0.682±0.008
R_3 (EnCodec loop)	-0.27±0.25	0.137±0.179	0.127±0.031	1.000±0.000	0.920±0.001	0.637±0.046
R_3^{multi} (multi-space)	0.95±0.16	0.082±0.107	0.016±0.026	0.841±0.089	0.914±0.003	0.552±0.015
noise pairs	-0.09±0.01	0.270±0.107	0.230±0.028	1.000±0.000	0.917±0.004	0.722±0.013
PGD pairs	-0.08±0.02	0.223±0.057	0.225±0.044	0.969±0.025	0.909±0.001	0.635±0.009
random-latent pairs	-0.03±0.03	0.019±0.016	0.200±0.053	1.000±0.000	0.922±0.004	0.696±0.010

TABLE III

DOWNSTREAM GRPO ENHANCEMENT. CHANGE AGAINST THE SUPERVISED PRE-TRAINED ENHANCER ON VOICEBANK-DEMAND TEST, PER REWARD PREDICTOR. MEAN±SD OVER 3 SEEDS (R_2: ONE SEED).

Measure	R_0	R_3	random-latent pairs
Δ own reward	0.263±0.010	0.269±0.028	0.334±0.051
ΔPESQ	-0.226±0.170	-0.034±0.126	0.019±0.050
ΔESTOI	-0.057±0.010	-0.038±0.013	-0.043±0.006
ΔSI-SDR (dB)	-6.1±1.4	-5.8±1.4	-5.7±0.6
ΔDNSMOS	-0.075±0.031	-0.076±0.056	-0.075±0.043
ΔNISQA	0.335±0.186	0.300±0.072	0.468±0.059
ΔUTMOS	0.018±0.006	0.057±0.030	0.076±0.011

and are not independent). A predictor taught that perturbed audio is bad seems to generalise that to genuinely noisy audio.

Downstream, patched predictors are safer rewards — and so is the baseline (Table III). Against R_0 the enhancer shows the textbook hack: reward $+0.263$ while PESQ falls -0.226, ESTOI -0.057 and SI-SDR -6.1 dB. Against R_3 the reward rises as much ($+0.269$) with PESQ intact (-0.034), and R_2 behaves the same ($+0.378$, -0.034), so this is a property of patching rather than of one round; the random-pair baseline again matches it ($+0.019$). Preregistered H3 is *not* supported (2.33 degraded measures for both rewards): NISQA and UTMOS *rise* and DNSMOS falls under every reward, so none detects this hack, while speaker similarity and WER barely move. R_3 scores the R_0-hacked enhancer 2.77 against 3.72 for the pre-trained model, flagging a generator that hacked a *different* predictor but not the one that hacked itself (3.99); without a loudness-matched control we cannot separate this from "scores degraded audio lower".

VI. CONCLUSION

How hackable a speech quality predictor is depends on two measurement choices the field has not standardised: whether the hack rule is referenced to the processing chain's own output, and whether hackability is reported for one attacker or the worst of several. Corrected on both, common predictors differ by more than an order of magnitude at one perturbation size — NISQA 0.90 against UTMOS 0.06 — so the choice of reward model is itself a robustness decision. Patching a predictor on its own hacks hardens only the space it trains on, and a random-perturbation baseline matches it.

Two caveats bound these conclusions. Our attacker is black-box and mild — its hacks against SSL-MOS move ESTOI by -0.023 and SI-SDR by -1.034 dB, and two independent judges call them perceptually marginal (UTMOS -0.004, DNSMOS -0.048; PGD moves UTMOS $+0.802$) — and we ran no listening test, so the benchmark ranks predictors against *this* attack family only. The detector is not fully independent either: two panel measures are neural, and for waveform attacks its reference is the input itself, so ASR is not comparable across families. The patch study uses $n = 3$ seeds and one predictor family (46 A100-hours).

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